Recall

Work from memory, then check with your tutor.

R1. 7×8

Answer: _____

R2. $96 \div 12$

Answer: _____

R3. $\frac{1}{2} + \frac{1}{4}$

Answer: _____

R4. $0.6 + 0.25$

Answer: _____

S. Return here near the end of the lesson

When I divide, I first ask whether I need the number of ___ or the size of ___.

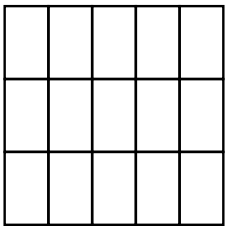
Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Five bags each hold $\frac{3}{4}$ cup of seeds. How many cups altogether?

Answer: _____

P2. A rectangle is $\frac{3}{5}$ meter long and $\frac{2}{3}$ meter wide. Find its area using equal parts.



P3. Find $\frac{2}{3}$ of $1\ \frac{1}{2}$ cups.

Answer: _____

Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Without multiplying first, compare $\frac{3}{4} \times 8$ and $\frac{5}{4} \times 8$ with 8. Then compute.

Answer: _____

P5. Three meters of ribbon are cut into $\frac{1}{4}$-meter pieces. How many pieces?

Answer: _____

P6. A $\frac{1}{4}$-liter juice sample is shared equally among three cups. How much goes in each cup?

Answer: _____

Fraction operations with models

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Write one story for $2 \div \frac{1}{5}$ and one for $\frac{1}{5} \div 2$. Solve both.

Answer: _____

F. A model has 4 columns and 3 rows. Shade $\frac{3}{4}$, then $\frac{2}{3}$ of it. Complete __ × __ = __.

Answer: _____

D. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

Answer: _____

Mixed review

M1. 13×4

Answer: _____

M2. $72 \div 9$

Answer: _____

M3. $\frac{1}{3} + \frac{1}{6}$

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A board is 5 meters long. Each shelf needs $\frac{1}{2}$ meter. How many shelves can be cut?

Answer: _____

Fractions and equal parts

The big idea

The whole stays the same when fractions are compared. Multiplication can take part of an amount. Division can count fractional groups or split a fraction into shares.

Words to know

Numerator: The top number counts equal parts. Example: In $\frac{3}{5}$, count three fifths.

Denominator: The bottom number names how many equal parts make one whole. Example: Five fifths make one whole.

Equivalent fractions: Different names for the same amount. Example: $\frac{1}{2} = \frac{3}{6}$.

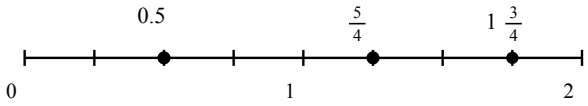
Quotient: The result of division. Example: $3 \div 4 = \frac{3}{4}$.

Scaling: Multiplying an amount to change its size. Example: $\frac{1}{2} \times 8$ takes half of eight.

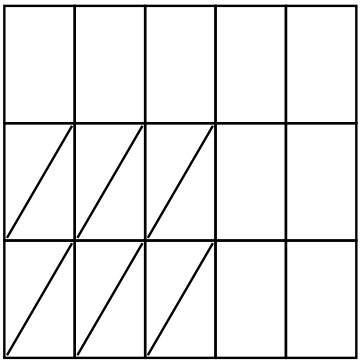
Unit fraction: A fraction with numerator 1. Example: $\frac{1}{5}$ is one fifth.

Product: The result of multiplication. Example: $\frac{1}{2} \times 8$ has product 4.

How it works



One whole keeps the same length. A fraction names equal parts of one whole. The size of that whole stays fixed.



$\frac{2}{3}$ of $\frac{3}{5}$ is $\frac{6}{15}$, or $\frac{2}{5}$. Multiplication can take part of an amount. Division can count groups or find the size of each share.

Worked examples

1. Compare $\frac{2}{3}$ and $\frac{3}{4}$.

Use one whole for both fractions.

Split the whole into 12 equal parts.

$\frac{2}{3} = \frac{8}{12}$; $\frac{3}{4} = \frac{9}{12}$.

Eight twelfths is less than nine twelfths.

Answer: $\frac{2}{3} < \frac{3}{4}$

2. Find $\frac{3}{4}$ of $\frac{2}{3}$ cup.

Split a whole into 3 equal columns and 4 equal rows.

Select 2 columns, then 3 fourths of those columns.

The overlap is 6 of 12 pieces: $\frac{6}{12} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ cup

Common mistakes

Places $\frac{5}{4}$ between 0 and 1. Why: Thinks all fractions are less than one. Fix: Build four fourths, then add one fourth.

Orders by denominator size. Why: Confuses number of parts with part size. Fix: Compare equal-length strips cut into fifths and ninths.

Says multiplication always increases an amount. Why: Uses only whole-number experience. Fix: Take half of a familiar amount.

Try it

1. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

2. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

3. Find $\frac{3}{4} \times \frac{4}{5}$.

4. Find $\frac{3}{5}$ of 1 $\frac{2}{3}$ cups.

5. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

Answers

1. $\frac{2}{9} < \frac{7}{9}$ 2. $\frac{7}{8}$ 3. $\frac{3}{5}$ 4. 1 cup 5. Smaller; 6

Where this connects

This builds on equal sharing and multiplication facts. Next, ratios compare amounts that change together.

Tape Diagram Template

